Before class: Windows and Mac setup

TiPS practical | Student installation guide

1. Prepare your computer a few days before class

The live practical needs **R, ape, TiPS 1.3.0 and C++ build tools**. RStudio is optional and does not supply the compiler. Installing the R packages alone is not sufficient: TiPS compiles your model when you run it. Installation is outside the 30-45 minute practical.

Install R from CRAN. On a Mac, choose the installer for your processor (Apple silicon or Intel). Unzip the **whole student archive** into a folder you can write to. Keep all files and subfolders together.

2. Install the build tools for your system

Windows

1. In the **R console**, type `R.version.string` to see your R version.
2. Open the official Rtools page and select the Rtools version listed for your R version and machine architecture. Use its installer and the standard installation location.
3. Close all R and RStudio windows, then reopen R/RStudio. Do not edit PATH manually as a first troubleshooting step.

macOS

1. Open **Terminal** from Applications > Utilities. This is not the R console.
2. Run the command below and complete the Apple installation dialog:

```
xcode-select --install
```

3. Restart R/RStudio after installation. The Command Line Tools provide the C/C++ compiler; the full Xcode application is not required for this TP. If macOS says they are already installed, continue to the actual check on page 2.

If installation of an R dependency from source reports a missing **gfortran**, ask the instructor or IT for the Fortran version compatible with your R release. Do not install a random compiler. See the official R for Mac tools page.

Managed or borrowed computer: if installation is blocked or requires administrator credentials you do not have, contact IT before class. The supplied scripts do not install system build tools automatically.

3. Open the student folder in R

In RStudio, open **TiPS_student.Rproj**. In plain R, set the working directory to the unzipped student folder, using your own path, for example:

```
setwd("C:/Users/yourname/Documents/TP_TiPS_ETUDIANTS") # Windows
# Mac example: setwd("/Users/yourname/Documents/TP_TiPS_ETUDIANTS")
file.exists("check_setup.R")  # Must return TRUE
```

4. Install packages, then test actual compilation

Run these commands **in the R console**, in this order:

```
source("00_install_packages.R")
source("check_setup.R")
```

The first script installs missing packages and dependencies in the local `.R-library` folder. Internet access is needed for downloads. If asked to compile a newer package from source when a binary is available, use the binary for classroom preparation.

The second script compiles a tiny TiPS model, runs one recovery event and checks its output. It does not run the whole TP. Your computer is ready only when you see:

```
READY: TiPS 1.3.0 compiled, simulated an event and retained its trace.
```

If the check fails

- **Missing packages:** rerun the installation script; keep its error message if installation fails.
- **Wrong TiPS version:** this teaching wrapper requires **1.3.0**. Ask the instructor to help install the tested version; a newer version is not automatically compatible.
- **C++ compilation:** follow the Windows or Mac advice printed by the check. Restart R after installing build tools and rerun the check. A compilation failure can also be a configuration or dependency problem, not just an absent compiler.
- **Mac tools already installed:** in Terminal run `xcode-select -p`. If this fails or compilation still fails, send the error to the instructor/IT; do not change system paths at random.
- **Simulation/output failure:** compilation worked, but the native test failed. Send the error to the instructor.

Keep the **full console error** and **setup_diagnostics.txt**, if created. The file records the failed stage, original R error and software versions; compiler details may appear only in the console. Send both to the instructor before class.

During class: start the practical or use explicit rescue

After READY, open **TiPS_student.R** and execute its five numbered blocks in order. Keep `use_precomputed <- FALSE` for live simulations.

If setup is still blocked, spend no more than two minutes troubleshooting during the session. For rescue, install/load **ape** if needed:

```
install.packages("ape", repos="https://cloud.r-project.org")
```

In block 1 of **TiPS_student.R**, change `use_precomputed <- FALSE` to `use_precomputed <- TRUE`, then execute all blocks from the beginning. This uses the supplied results and needs neither TiPS nor a compiler. Figures are labelled **PRECOMPUTED RESCUE**. If R cannot run, use the supplied **figures** and **data** folders.

Terminal alternative (from the student folder): `Rscript run_without_rstudio.R` for live simulation; add `--rescue` to use the supplied results.

Setup has been executed on Linux. Windows/Mac installation guidance follows the official sources linked above; the full TP still needs a trial on those classroom systems.